TECHNICAL ARCHITECTURE DOCUMENT

# Salesforce CRM Platform

Northwind Bank · Sales Cloud · Financial Services Cloud · OmniStudio

| REFERENCE | VERSION | DATE |
| --- | --- | --- |
| TAD-NWB-CRM | 1.0 — For review | 28 October 2026 |

| PREPARED BY | RECIPIENT | CLASSIFICATION |
| --- | --- | --- |
| Acme Integrators | Northwind Bank | Confidential — fictional example |

Document prepared by Acme Integrators for Northwind Bank. Restricted distribution Northwind Bank / Acme Integrators.

# Document control

| Item | Value |
|---|---|
| Title | Technical Architecture Document — Salesforce CRM Platform |
| Reference | TAD-NWB-CRM |
| Client | Northwind Bank |
| Integrator | Acme Integrators |
| Version | 1.0 — For review |
| Date | 28 October 2026 |
| Classification | Confidential — fictional example |
| Scope | Salesforce organization (Production) of Northwind Bank, state as of 28 October 2026 |

## Version history

| Version | Date | Author | Purpose |
|---|---|---|---|
| 1.0 | 28 October 2026 | Acme Integrators | Consolidated version submitted to Northwind Bank for review and approval |

# 1 Executive summary

The **Northwind Bank CRM platform** is the bank's system of engagement for retail and business customers. Designed and built by Acme Integrators on Salesforce, it supports two populations, each in a dedicated console application:

- the **branch network** (advisors and branch managers) in the « Branch Console », to manage customers, leads and loan applications;
- the **contact center** (agents and their manager) in the « Contact Center » console, to qualify leads and record contact attempts.

Customers interact indirectly through the **mobile banking app**, which creates leads through an inbound API.

The solution relies on four Salesforce foundations, complemented by specific developments:

- **Sales Cloud** carries the lead and opportunity lifecycle, from the first contact to the loan;
- **Financial Services Cloud** provides the financial account model that loan applications refer to;
- **OmniStudio** guides the loan intake journey;
- **Platform Events** decouple the loan decision from its follow-up actions.

**2**
Lightning applications

**4**
Custom business objects

**12**
Apex application classes, 5 test classes

**4**
Lightning Web Components

**1 / 1**
OmniScripts / Integration Procedures (active)

**3**
Active flows

**5**
Custom Metadata records (2 types)

**6**
Connected systems

## 1.1 Business capabilities covered

| Domain | Main capabilities |
|---|---|
| Customer knowledge | Customer record with live positions from the core banking system (Customer 360), segment kept in sync every night |
| Leads | Leads from the mobile app, the branches and the contact center; default branch assignment; scoring at creation |
| Loan applications | Intake guided by an OmniScript, automatic scoring from configurable rules, decision, document archiving |

| Domain | Main capabilities |
|---|---|
| Customer notifications | SMS reminders for missing documents, follow-up tasks when a status changes |
| Contact center | Lead queue, contact attempts and their outcome |
| Steering | Loan reports by status and by branch, loan pipeline dashboard |
| Security | Single sign-on for employees, persona-based permission set groups, role hierarchy following the network |

## 1.2 Key achievements

### Configurable loan scoring

The score of each loan application is computed by `LoanScoringService` from rules stored as `ScoringRule__mdt` records. A new criterion or a new weight is a configuration change: no code, no deployment window.

### Live Customer 360

Accounts, cards and loans are read in real time from the core banking system through a named credential ( `NWB_CoreBanking` ). No position is copied into Salesforce, so advisors always see the figures of the system of record.

### Event-driven loan follow-up

Every status change of a loan application publishes a `LoanStatusChange__e` event. Follow-up tasks, notifications and archiving subscribe to it, so new reactions are added without touching the trigger.

### Secure by design integrations

All calls go through named and external credentials; the inbound lead API runs in user mode; employees log in once through the corporate identity provider, with users created by a registration handler.

## 1.3 Architecture principles applied

1. **Standard first**: Sales Cloud objects, Financial Services Cloud accounts, OmniStudio journeys, queues and platform events are used before any specific development.
2. **The core banking system stays the master** of customers and positions: Salesforce reads them live and keeps only what the relationship needs.
3. **Configuration in Custom Metadata**: scoring rules and integration timeouts are records, changed without code.
4. **One trigger per object**, delegating to a handler class; services and selectors separate business logic from data access.
5. **Asynchronous for anything slow**: archiving, SMS and nightly synchronisation run outside the user transaction.
6. **Credentials outside the code**: every endpoint is a named credential; authentication is held by external credentials.

# 2 Purpose, scope and references

## 2.1 Purpose of the document

This document describes the technical architecture of the Northwind Bank CRM platform as implemented. It is the reference for:

- the understanding of the solution by the bank's teams;
- run, corrective and evolutive maintenance;
- architecture and security reviews;
- knowledge transfer between Acme Integrators, Northwind Bank and its partners.

It is written for architects, application owners, operations and security teams, and the developers who will make the platform evolve.

## 2.2 Scope

| Included | Excluded |
|---|---|
| The Salesforce organization of Northwind Bank (production): data model, application components, automation, OmniStudio, user interface, integrations seen from Salesforce, access model, delivery and operations | Business data; internals of the core banking system, the SMS gateway, the document store, the identity provider and the mobile app; internals of the Financial Services Cloud and OmniStudio packages |

## 2.3 References

| Ref. | Document | Purpose |
|---|---|---|
| DR-01 | Metadata of the Northwind Bank organization, retrieved in read-only mode | Source of every inventory and figure of this document |
| DR-02 | Salesforce Well-Architected (architect.salesforce.com) | Reading grid of chapter 13 |
| DR-03 | Salesforce Integration Patterns and Practices | Classification of the integration flows |

> **NOTE**
>
> Northwind Bank and Acme Integrators are fictional. This example shows what TADForce produces on a small org.

## 2.4 Method

This document describes the architecture **as implemented** in the Salesforce organization of Northwind Bank (Production), as of 28 October 2026. It was produced from the metadata of the organization, retrieved in read-

only mode, and from the inputs provided by Northwind Bank and Acme Integrators.

- Inventories and counts cover the components built for Northwind Bank. Components delivered by managed packages and unused standard samples are excluded.
- Each statement relies on the metadata, on a document provided by the project, or is explicitly marked as to be confirmed.
- Authentication parameters (identifiers, secrets, certificates) and business data are never reproduced.

The following components exist in the organization but are outside the documented architecture:

| Type | API name | Reason |
|---|---|---|
| apex | `DemoDataLoader` | Demonstration data loader |
| flow | `Old_Branch_Survey` | Draft survey never activated |
| lwc | `demoCarousel` | Demonstration component, not placed in production pages |

## 2.5 Reading conventions

- Technical names (objects, fields, classes, flows, components) are written in monospace, for example `Account`. Business labels are written in quotes, for example « Customer ».
- Salesforce terms in common use (Lead, Queue, Record Type, Permission Set, Named Credential…) are kept as is.
- Diagrams share one colour code:

Actors: users, populations, channels

Salesforce components

Systems outside the CRM

Data: objects, metadata, reference data

# 3 Context and vision

## 3.1 Business drivers

- Give every advisor a complete and current view of the customer, without leaving the CRM.
- Turn every contact — branch, contact center, mobile app — into a qualified lead handled by the right branch.
- Shorten the time from loan request to decision, with a score computed the same way everywhere.
- Keep customers informed at each step of their loan application.
- Steer the loan pipeline by status and by branch.

## 3.2 Actors and user populations

| Population | Application | Main activities |
|---|---|---|
| Branch advisors | « Branch Console » | Customer 360, leads of the branch, loan applications, contact attempts |
| Branch managers | « Branch Console » | Same as advisors, plus branch management and review of submitted loan applications |
| Contact center agents | « Contact Center » | Lead qualification from the contact center queue, contact attempts |
| Mobile app customers | Mobile banking app | Request a call back or a loan: a lead is created through the API |

## 3.3 Functional scope

| Domain | Covered by |
|---|---|
| Leads and qualification | § 5.2 |
| Loan applications | § 5.3 |
| Customer 360 | § 5.4 |
| Notifications | § 5.5 |
| Contact center | § 5.6 |

## 3.4 Key figures

| Area | Count |
|---|---|
| Lightning apps | 2 |
| Lightning pages | 3 |
| Lightning Web Components | 4 |
| Apex classes | 12 |

| Area | Count |
|---|---|
| Apex test classes | 5 |
| Apex triggers | 2 |
| Flows (active / total) | 3 / 3 |
| OmniScripts (active) | 1 |
| Integration Procedures (active) | 1 |
| Data Mappers | 1 |
| Custom objects | 4 |
| Custom fields | 30 |
| Custom Metadata types / records | 2 / 5 |
| Permission set groups / permission sets | 3 / 5 |
| Roles | 10 |
| Named / External credentials | 3 / 2 |
| Apex REST services | 1 |
| Reports / dashboards | 2 / 1 |

# 4 General architecture

## 4.1 Context view

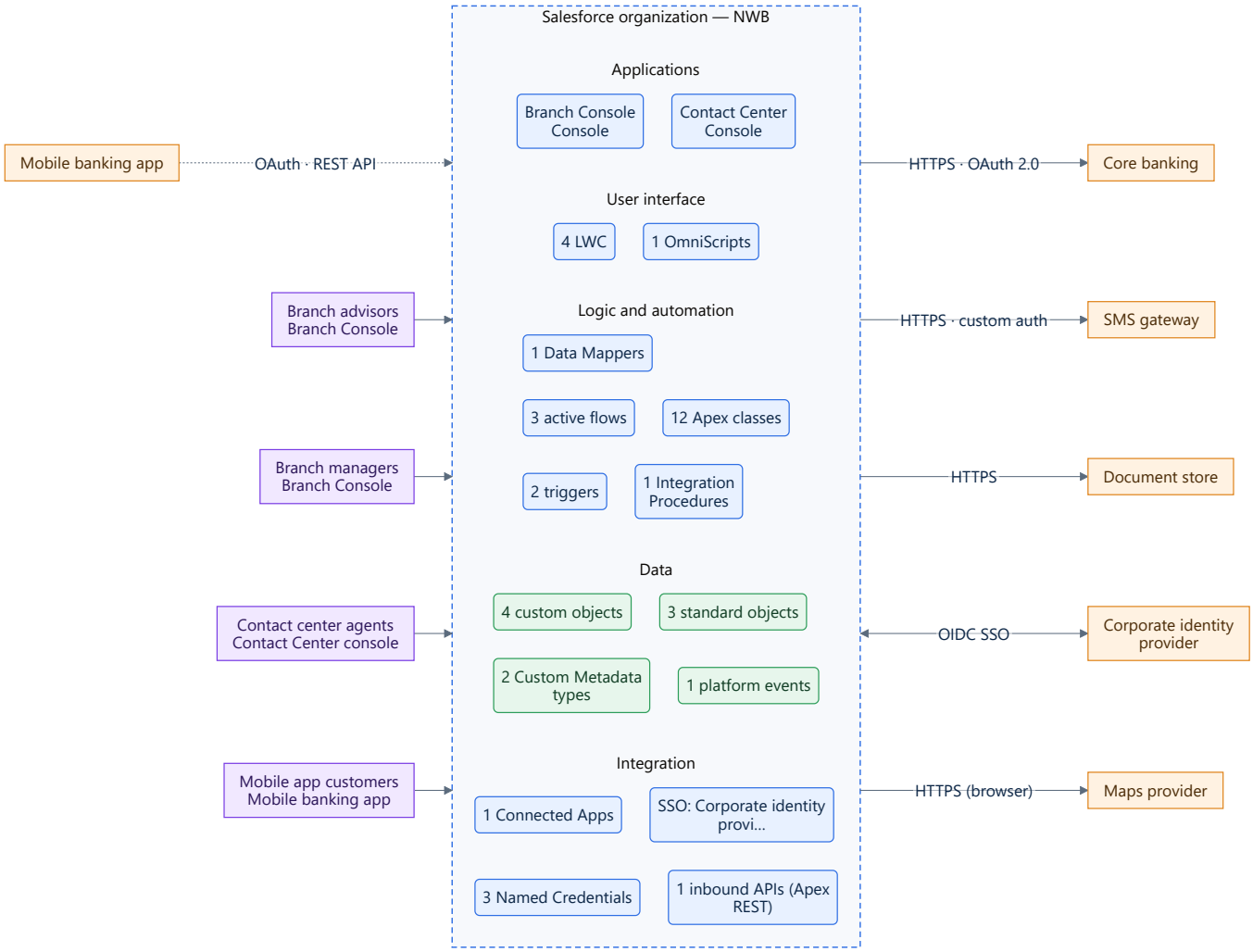



*Figure 1 — Context view: users, Salesforce organization and connected systems*

Actors: users, populations, channels | Salesforce components | Systems outside the CRM | Data: objects, metadata, reference data

**Users.** Employees reach the platform through two Lightning console applications and log in with the corporate identity provider. Customers never log in to Salesforce: the mobile banking app calls an Apex REST service on their behalf.

**Salesforce.** The organization combines Sales Cloud objects, the financial account model of Financial Services Cloud, an OmniStudio journey for loan intake and platform events for the loan follow-up. Specific objects hold branches, loan applications, loan documents and contact attempts.

**Connected systems.** The core banking system is read in real time and feeds the nightly customer synchronisation; the SMS gateway sends reminders; the document store archives the documents of decided loan applications; the maps provider serves map tiles to the console.

## 4.2 Salesforce foundation and products

| Product | Evidence in the metadata |
|---|---|
| Sales Cloud | object `Opportunity`, object `Lead` |
| Financial Services Cloud | namespace `FinServ` |
| OmniStudio | dedicated metadata |
| Platform Events / Change Data Capture | dedicated metadata |

| Product | Use in the solution |
|---|---|
| Sales Cloud | Leads, conversion, opportunities linked to loan applications |
| Financial Services Cloud | Financial account model referenced by loan applications ( `LoanApplication__c.FinancialAccount__c` ) |
| OmniStudio | Guided loan intake (OmniScript), server-side retrieval of positions (Integration Procedure), loan data extraction (Data Mapper) |
| Platform Events | Decoupling of the loan status change from its consequences |

Everything else — branches, loan applications, documents, scoring, Customer 360 — is specific and documented in chapters 6 to 9.

## 4.3 Logical architecture in layers

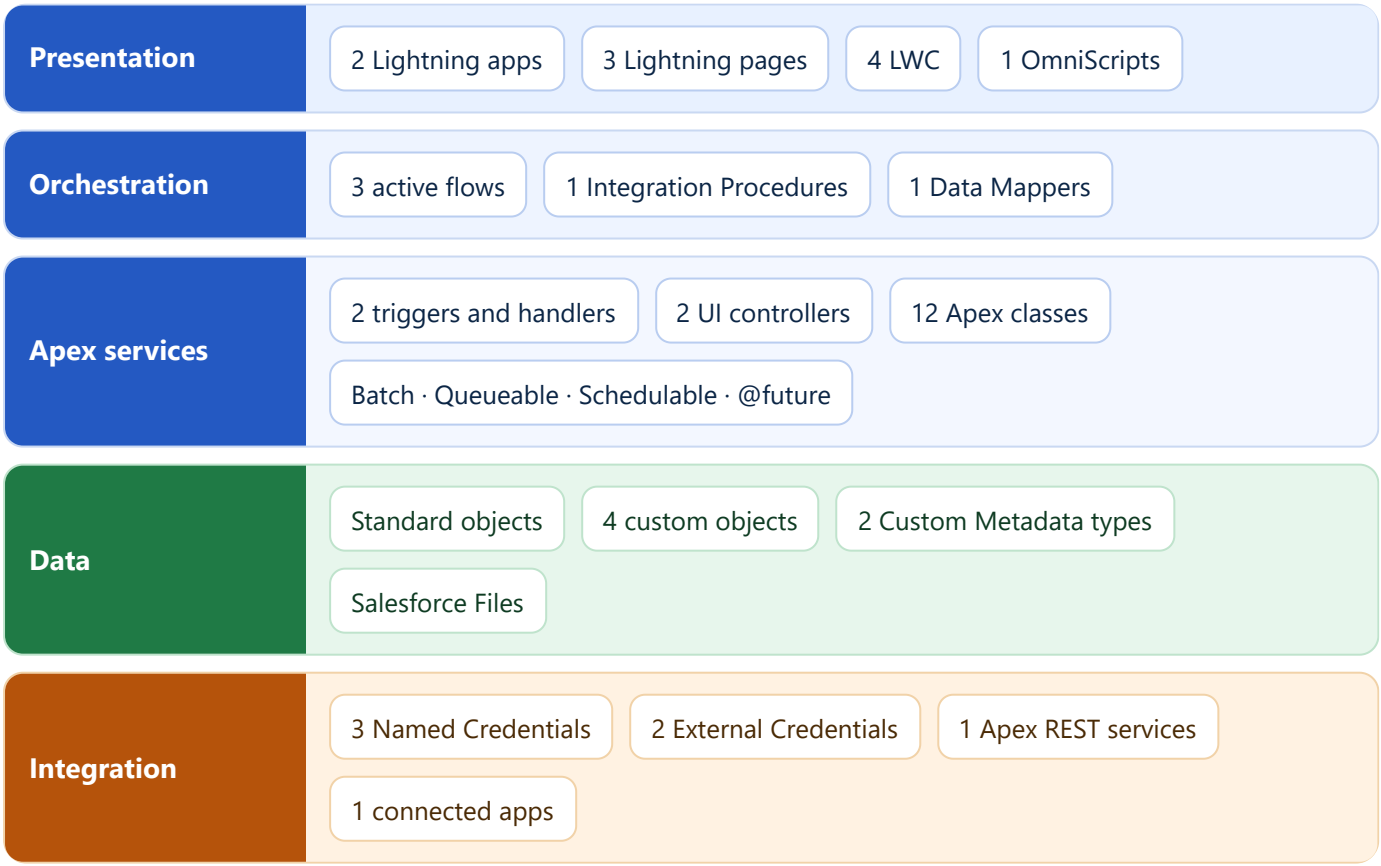



*Figure 2 — Logical architecture in layers*

| Layer | Components | Responsibility |
|---|---|---|
| Presentation | Console applications, Lightning pages, LWC, OmniScript | Show information and guide the user; no heavy business rule in the browser |
| Orchestration | Flows, Integration Procedure, Data Mapper | Chain the steps of a process; declarative automation |
| Apex services | Triggers and handlers, controllers, services, selectors, asynchronous jobs | Business rules, data access, integrations, volume processing |
| Data | Standard and industry objects, specific objects, Custom Metadata | Business data and configuration |
| Integration | Named and external credentials, Apex REST, connected app | Isolate and secure the exchanges with the bank's systems |

## 4.4 Design principles

- **Standard first**: standard objects, queues, record types, platform events and OmniStudio before custom code.
- **Configuration over code**: business parameters in Custom Metadata (`ScoringRule__mdt`, `IntegrationSetting__mdt`).
- **Bulk-safe**: triggers process lists, the customer synchronisation (`CustomerSyncBatch`, designed to run nightly) works in batches of 50 records.
- **Secure server-side**: controllers run `with sharing` and query in user mode; the inbound API inserts records as the calling user.
- **Loose coupling**: the trigger publishes an event; subscribers react independently.

## 4.5 Naming conventions

| Family | Observed convention | Occurrences |
|---|---|---|
| Apex | *Test | 5 |
| Apex | *Service | 2 |
| Apex | *TriggerHandler | 2 |
| Apex | *Batch | 1 |
| Apex | *Client | 1 |
| Apex | *Controller | 1 |
| Apex | *Handler | 1 |
| Apex | *Helper | 1 |
| Apex | *Queueable | 1 |
| Apex | *Resource | 1 |
| Flows | Loan_* | 2 |
| PermissionSets | NWB_* | 5 |

- Apex classes carry their role as a suffix: `*TriggerHandler`, `*Service`, `*Selector`, `*Controller`, `*Batch`, `*Queueable`, `*Client`, `*Resource`; test classes end with `Test`.
- Permission sets and permission set groups start with the bank code `NWB_`; persona groups with `NWB_Persona_`.
- Flows start with the object they work on (`Lead_`, `Loan_`).
- LWC names are in camel case and describe the component (`customer360`, `loanTimeline`).

# 5 Functional architecture and business processes

## 5.1 Functional domains

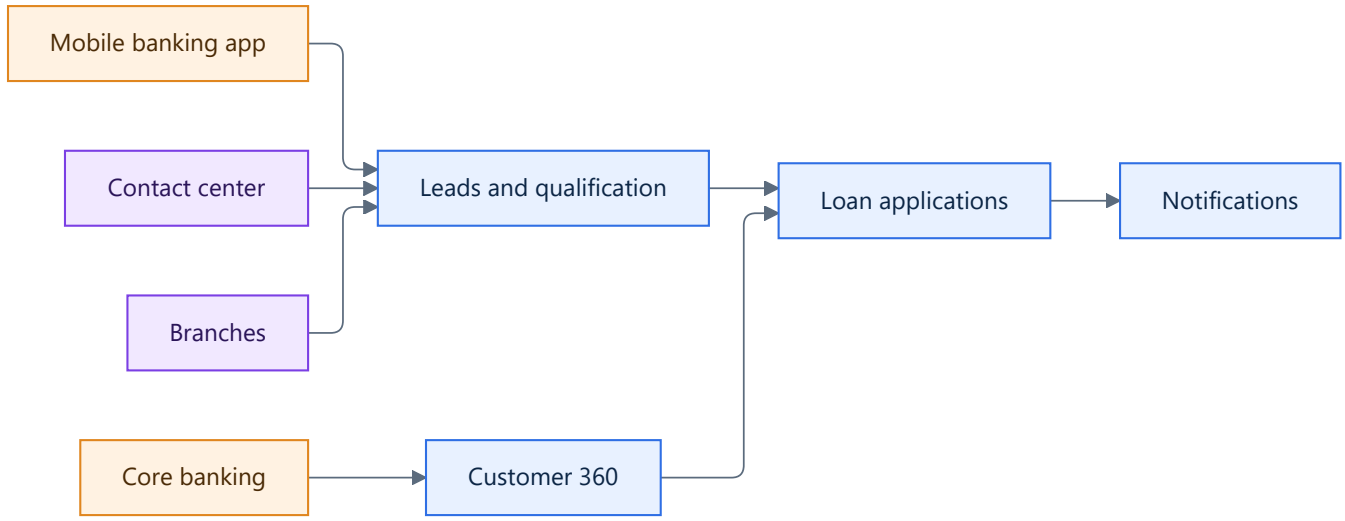



*Figure 3 — Functional domains and their sequence, from the first contact to the loan*

| Domain | Main objects | Main components | Section |
|---|---|---|---|
| Leads and qualification | `Lead`, `ContactAttempt__c` | `LeadApiResource`, `LeadTrigger`, `Lead_Assign_Branch` | § 5.2 |
| Loan applications | `LoanApplication__c`, `LoanDocument__c` | `NWB_LoanIntake_English_1`, `LoanApplicationTrigger`, `LoanScoringService` | § 5.3 |
| Customer 360 | `Account` | `customer360`, `Customer360Controller`, `CoreBankingClient` | § 5.4 |
| Notifications | `LoanStatusChange__e` | `Loan_Status_Notification`, `Loan_Document_Reminder`, `SmsNotificationService` | § 5.5 |
| Contact center | `Lead`, `ContactAttempt__c` | Queue `Contact_Center_Leads`, « Contact Center » console | § 5.6 |

## 5.2 Leads and qualification

Leads come from three channels: the mobile banking app, the branches and the contact center.

- The mobile app calls the Apex REST service `LeadApiResource` (`/leads/v1/*`). The lead is inserted in user mode with its channel (`Channel__c`).
- At creation, `LeadTrigger` delegates to `LeadTriggerHandler`, which initialises the score (`Score__c`).
- The record-triggered flow `Lead_Assign_Branch` then assigns the lead to its branch (`Branch__c`).
- Leads waiting for the contact center are owned by the queue `Contact_Center_Leads`.
- A converted lead can no longer change status: the validation rule `Lead_Locked_When_Converted` enforces it.

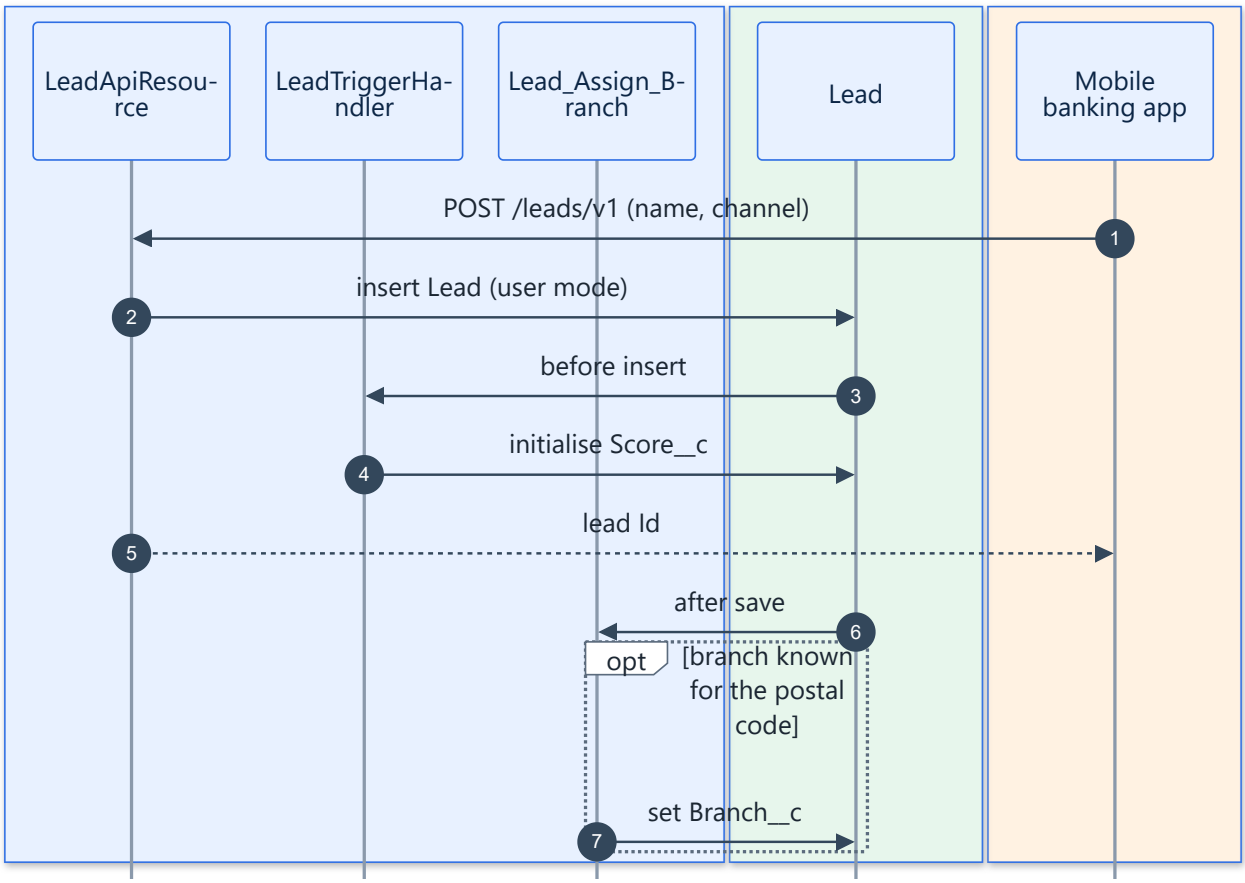



*Figure 4 — Sequence — Creation of a lead from the mobile banking app*

## 5.3 Loan applications

A loan application ( `LoanApplication__c` ) is created from the customer record (quick action `Account.New_Loan_Application` ), from the « New loan » tab ( `loanWizard` ) or through the guided OmniScript `NWB_LoanIntake_English_1` . Its record type distinguishes consumer loans from mortgages.

- **Scoring.** Before each save, `LoanApplicationTriggerHandler` asks `LoanScoringService` to compute `Score__c` from the `ScoringRule__mdt` rules.
- **Review.** Submitted applications are shared in read-only mode with the branch managers (sharing rule `Share_Submitted_With_Review_Team` ); the queue `Loan_Review` holds the applications waiting for a reviewer.
- **Decision.** Each status change publishes a `LoanStatusChange__e` event. When the status becomes « Approved » or « Rejected », the documents are archived by `DocumentArchiveQueueable` .
- **Documents.** Each document is a `LoanDocument__c` record, child of its application (master-detail), with its archive identifier ( `ArchiveId__c` ).

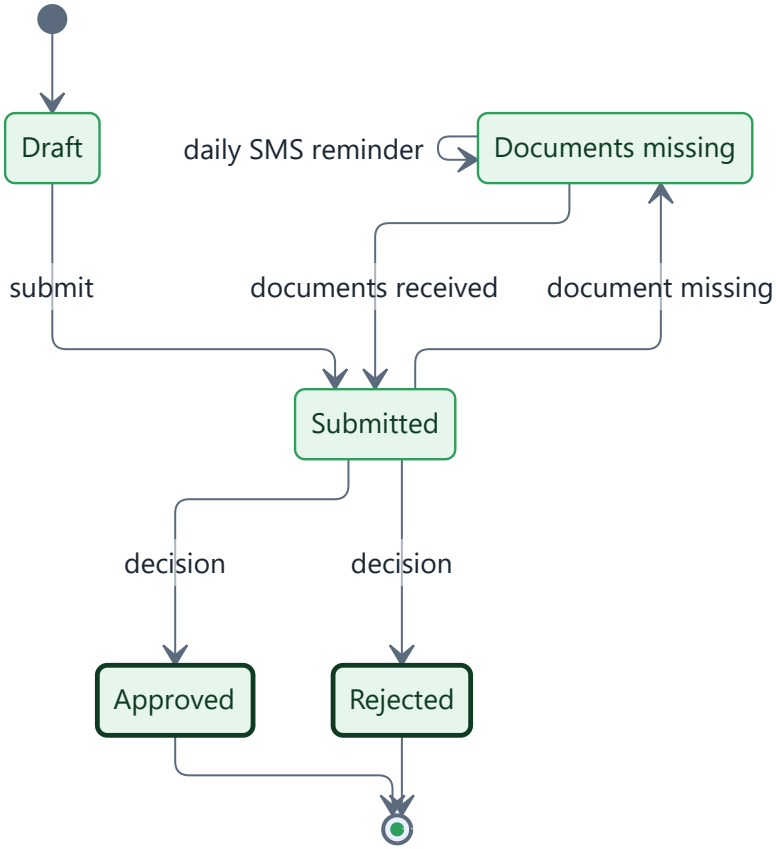



*Figure 5 — Statuses of a loan application (values to be confirmed with the business team)*

> **ACME INTEGRATORS CONTRIBUTION**
>
> Scoring rules are data, not code: the business team adjusts weights and criteria in `ScoringRule__mdt` without a release.

## 5.4 Customer 360

On the customer record page ( `Account_Record_Page` ), the `customer360` component displays the customer's positions read live from the core banking system:

1. the component calls `Customer360Controller.getPositions` through a cached wire adapter;
2. the controller reads the customer number ( `NWB_CustomerNumber__c` ) in user mode;
3. `CoreBankingClient` calls the core banking API through the named credential `NWB_CoreBanking` ;
4. each position is rendered by `positionTile` .

`CustomerSyncBatch` , designed to run each night once scheduled, refreshes the customer segment ( `Segment__c` ) from the same API.

## 5.5 Notifications

- **Status change.** The flow `Loan_Status_Notification` subscribes to `LoanStatusChange__e` and creates a follow-up task for the owner.
- **Missing documents.** The scheduled flow `Loan_Document_Reminder` runs daily on loan applications and calls the invocable action of `SmsNotificationService`, which sends the SMS through the SMS gateway asynchronously.

## 5.6 Contact center

Agents work in the « Contact Center » console on the leads of the queue `Contact_Center_Leads`. Each call is recorded as a contact attempt ( `ContactAttempt__c` ) linked to the lead or the customer, with its outcome ( `Outcome__c` ).

# 6 Data architecture

## 6.1 Modeling principles

- Standard objects carry the relationship: `Account` for customers (retail and business record types), `Lead`, `Opportunity`.
- Financial accounts follow the Financial Services Cloud model; loan applications refer to them.
- Specific objects exist only for what the standard does not cover: branches, loan applications and their documents, contact attempts.
- Every record fed by or sent to another system carries an external ID used as matching key.
- Relationships are lookups, except the documents of a loan application (master-detail: they live and die with it).
- Business parameters are Custom Metadata records.

## 6.2 Overview

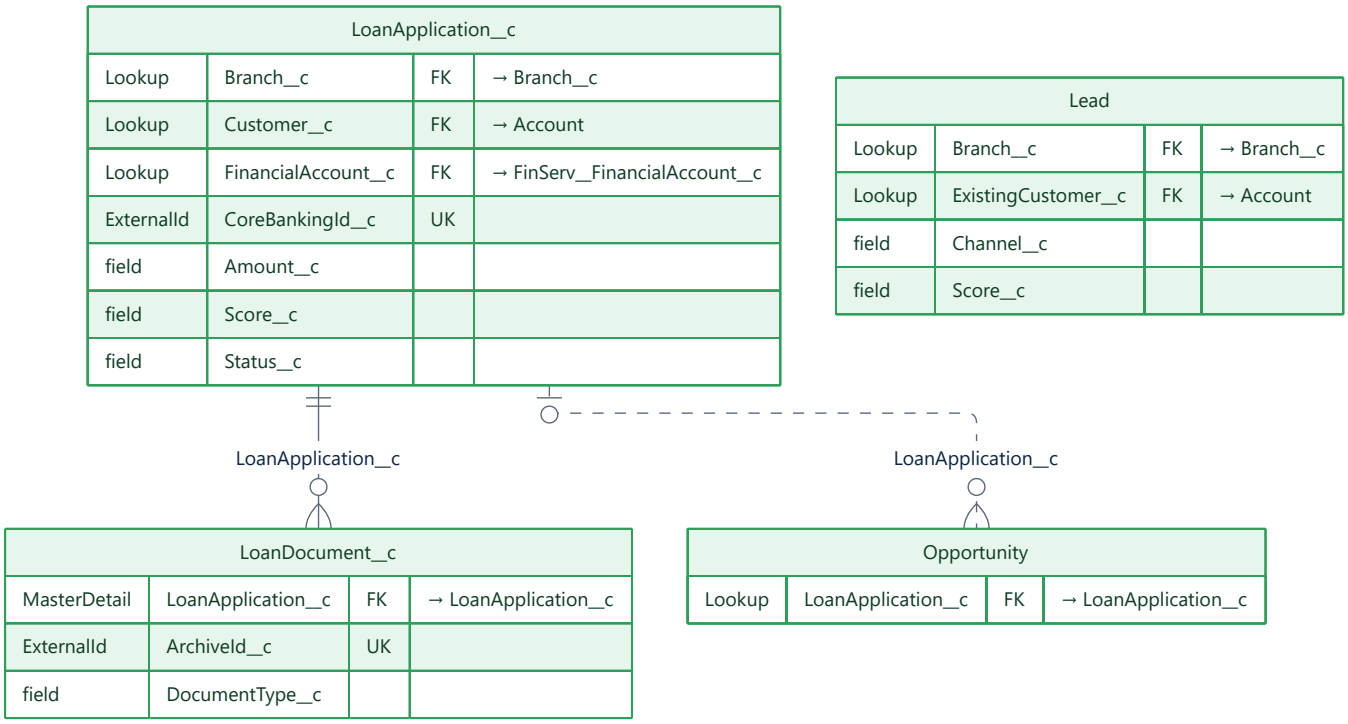



*Figure 6 — Main objects and their relationships (1/2)*

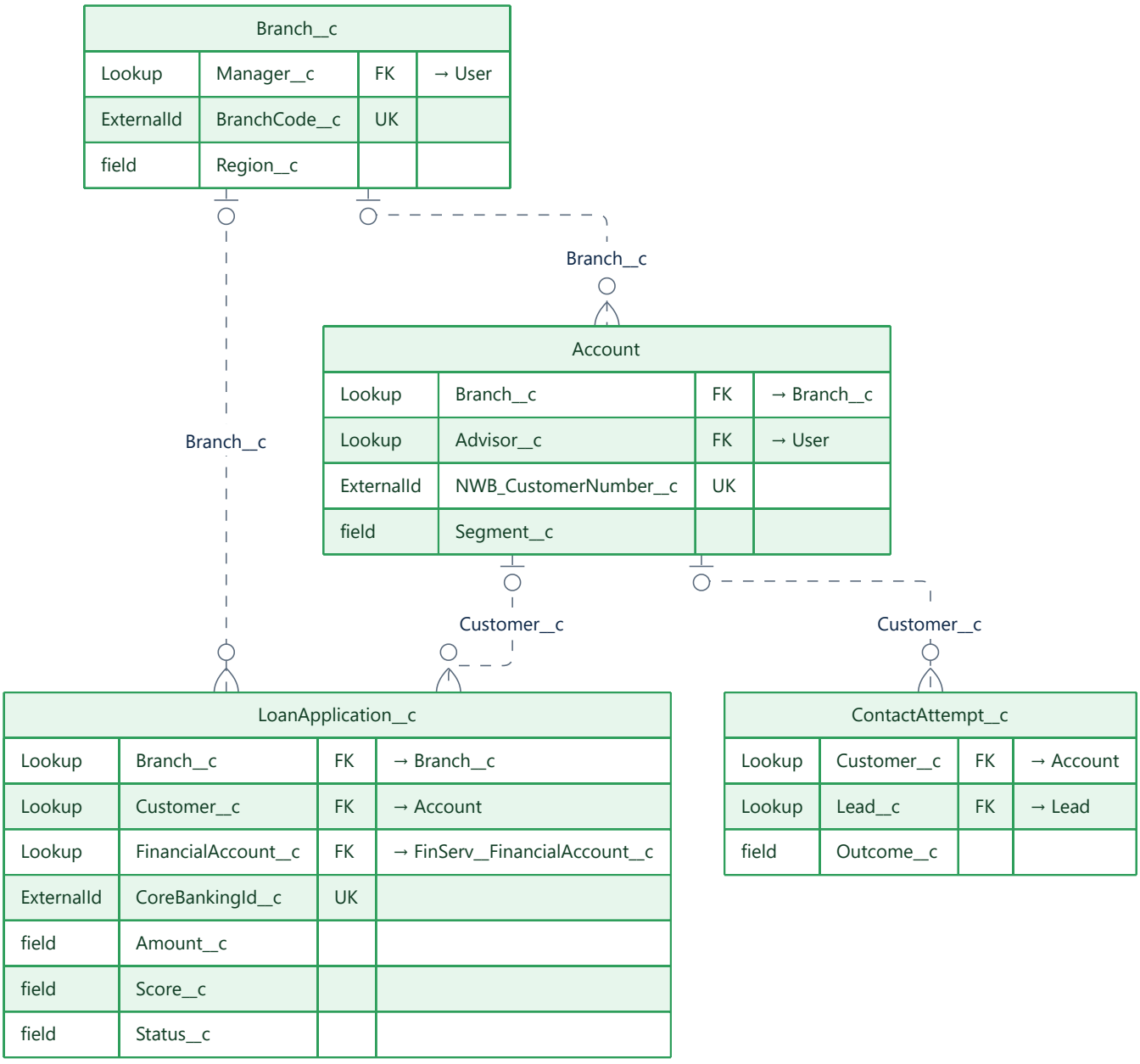



*Figure 7 — Main objects and their relationships (2/2)*

The customer ( `Account` ) is the centre of the model: its branch, its loan applications, its leads and its contact attempts hang off it. The branch ( `Branch__c` ) owns customers and leads by default. A loan application gathers its documents and may be linked to an opportunity and to a financial account.

## 6.3 Objects by domain

### 6.3.1 Customers and branches

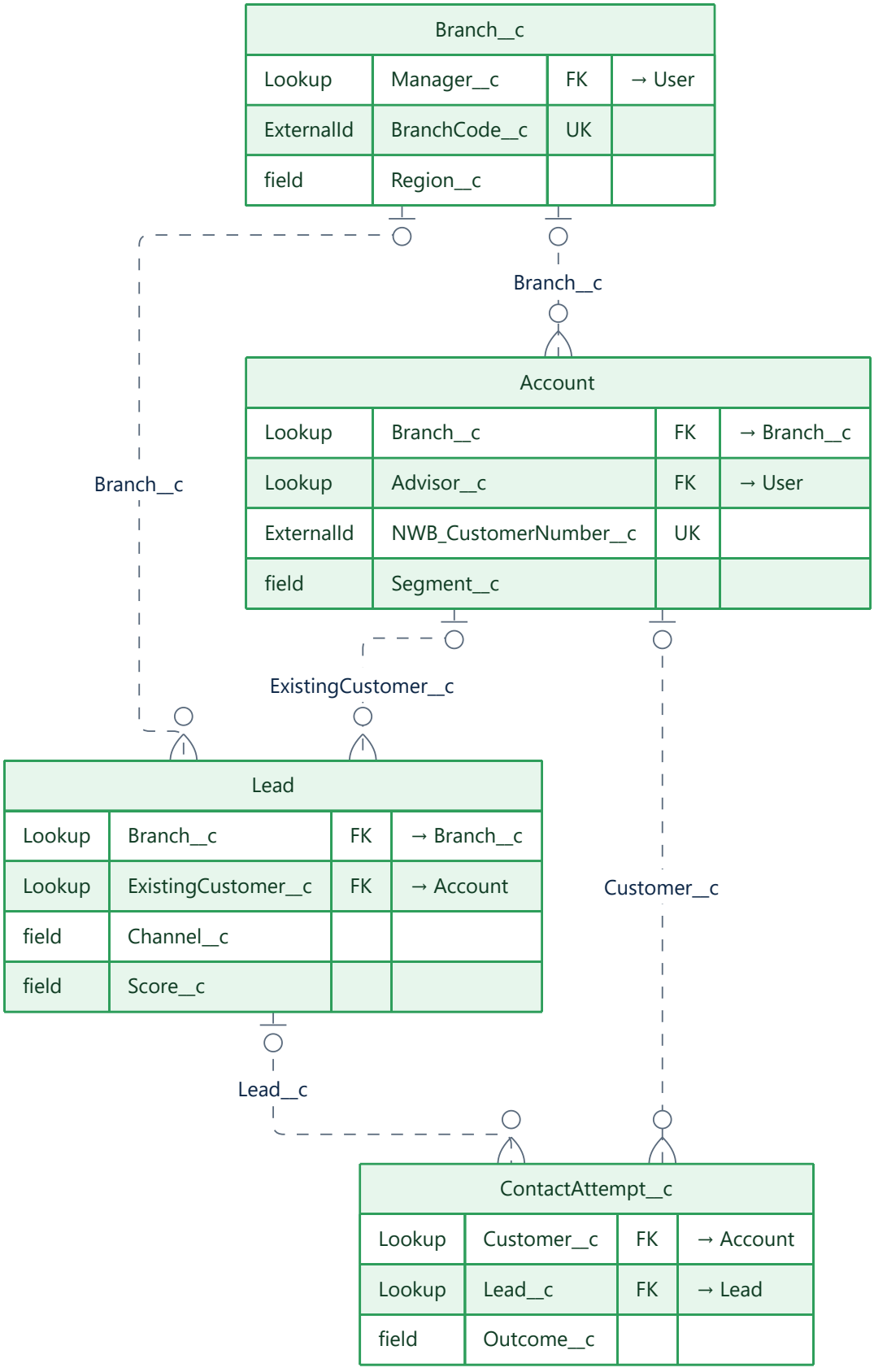



*Figure 8 — Data model — Customers, branches and leads*

| Object | Role | Key fields |
| --- | --- | --- |
| `Account` | Customer (retail or business) | `NWB_CustomerNumber__c` (core banking key), `Segment__c`, `Branch__c`, `Advisor__c` |
| `Branch__c` | Branch of the network | `BranchCode__c`, `Region__c`, `Manager__c` |
| `Lead` | Prospect or customer request | `Channel__c`, `Score__c`, `ExistingCustomer__c`, `Branch__c` |
| `ContactAttempt__c` | Call or contact by an advisor or an agent | `Customer__c`, `Lead__c`, `Outcome__c` |

### 6.3.2 Loans

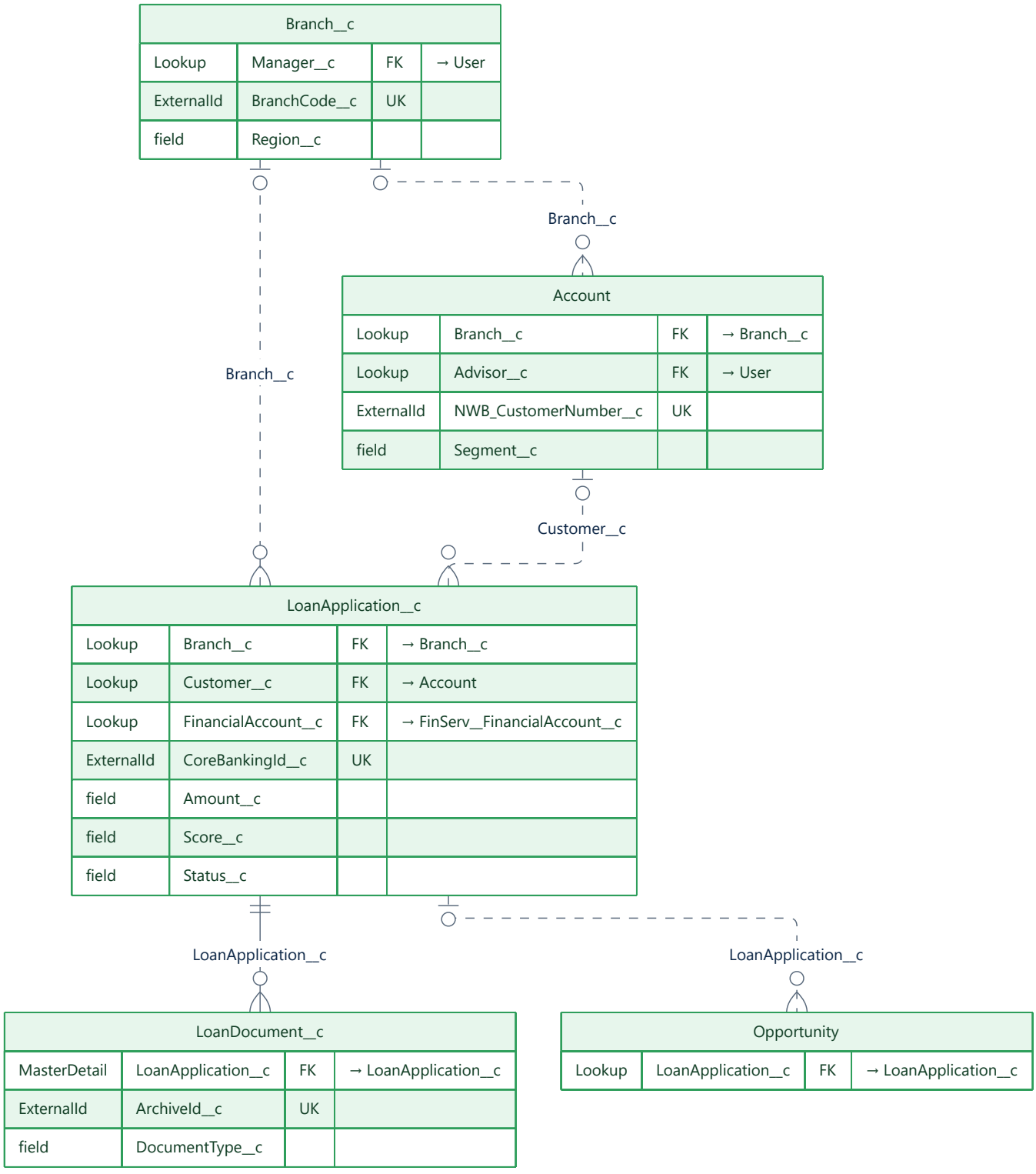



*Figure 9 — Data model — Loan applications and their documents*

| Object | Role | Key fields |
|---|---|---|
| `LoanApplication__c` | Loan request from intake to decision | `Customer__c`, `Status__c`, `Amount__c`, `Score__c`, `CoreBankingId__c`, `FinancialAccount__c` |
| `LoanDocument__c` | Document of an application | `LoanApplication__c` (master), `DocumentType__c`, `ArchiveId__c` |

| Object | Role | Key fields |
|---|---|---|
| `Opportunity` | Commercial opportunity of the loan | `LoanApplication__c` |

## 6.4 Industry cloud objects

| Object | Use |
|---|---|
| `FinServ__FinancialAccount__c` | Financial account (loan, current account) to which a loan application can be attached through `FinancialAccount__c` |

The rest of the Financial Services Cloud model is not referenced by the specific metadata.

## 6.5 External IDs and integration keys

| Object | Field |
|---|---|
| `Account` | `NWB_CustomerNumber__c` |
| `Branch__c` | `BranchCode__c` |
| `LoanApplication__c` | `CoreBankingId__c` |
| `LoanDocument__c` | `ArchiveId__c` |

- `NWB_CustomerNumber__c` and `CoreBankingId__c` are the keys of the core banking system, used by the nightly synchronisation and by the Customer 360 calls.
- `BranchCode__c` identifies a branch in the bank's referential.
- `ArchiveId__c` is the identifier returned by the document store.

## 6.6 Record types and business processes

| Object | Record types |
|---|---|
| `Account` | `Business_Customer`, `Retail_Customer` |
| `LoanApplication__c` | `Consumer_Loan`, `Mortgage` |

Customer record types separate retail and business customers (different page layouts and segments). Loan record types separate consumer loans and mortgages, which follow different scoring rules and documents.

## 6.7 Configuration in Custom Metadata

| Type | Records |
|---|---|
| `IntegrationSetting__mdt` | 2 |
| `ScoringRule__mdt` | 3 |

| Type | What it configures |
| --- | --- |
| `ScoringRule__mdt` | Criteria and weights of the loan score ( `Criterion__c` , `Weight__c` ) |
| `IntegrationSetting__mdt` | Timeouts of the outbound integrations ( `TimeoutMs__c` ) |

## 6.8 Platform events and change data capture

| API name | Label |
| --- | --- |
| `LoanStatusChange__e` | Loan status change |

`LoanStatusChange__e` carries the loan application identifier and its new status. It is published by `LoanApplicationTriggerHandler` and consumed by the flow `Loan_Status_Notification`. No change data capture entity is selected.

## 6.9 Files and documents

The documents of a loan application are described by `LoanDocument__c` records; their content is archived in the document store once the application is decided, and the archive identifier is kept on the record.

## 6.10 Data lifecycle

Field history tracking is enabled on the specific objects. No archiving or purge policy is configured in Salesforce for loan applications and contact attempts: the retention rules are to be confirmed with the bank.

# 7 Application architecture

## 7.1 Apex code architecture

| Nature | Count |
|---|---|
| Batch | 1 |
| Queueable | 1 |
| Schedulable | 1 |
| @future | 1 |
| REST | 1 |
| Invocable | 1 |
| LWC/Aura controller | 2 |
| Callout | 3 |
| SSO registration handler | 1 |
| Global | 2 |

### 7.1.1 Design principles

- One trigger per object, with no logic in the trigger body: each trigger calls the `run()` method of its handler.
- Business rules live in services ( `LoanScoringService` ), data access in selectors ( `LoanApplicationSelector` ), external calls in clients ( `CoreBankingClient` ).
- Code processes lists of records, never one record at a time.
- 8 classes declare `with sharing` ; queries from user-facing code run `WITH USER_MODE` .

### 7.1.2 Layers and conventions

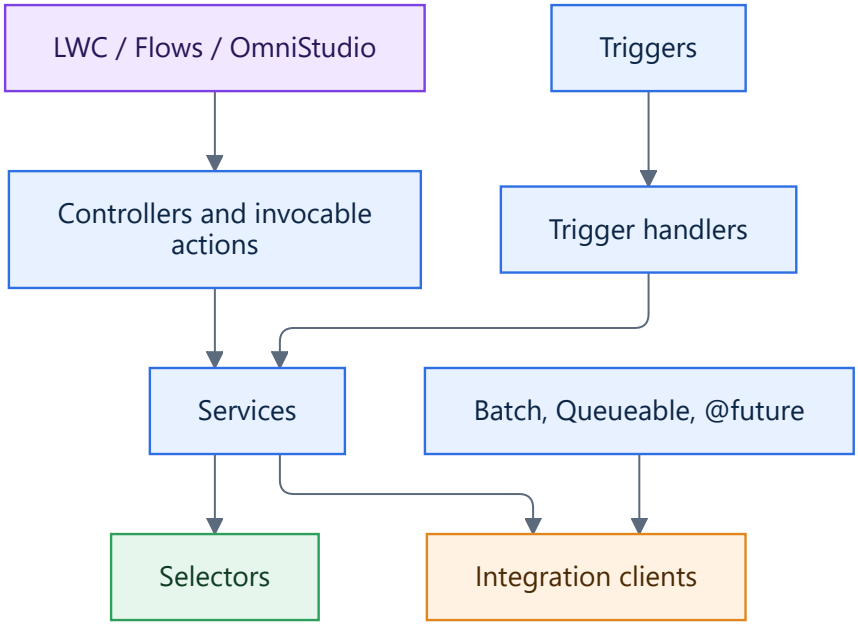



*Figure 10 — Layers of the Apex code*

| Layer | Convention | Examples |
|---|---|---|
| Trigger handlers | `<Object>TriggerHandler` | `LoanApplicationTriggerHandler`, `LeadTriggerHandler` |
| Services | `<Domain>Service` | `LoanScoringService`, `SmsNotificationService` |
| Selectors | `<Object>Selector` | `LoanApplicationSelector` |
| Controllers | `<Component>Controller` | `Customer360Controller` |
| Integration clients | `<System>Client` | `CoreBankingClient` |
| Asynchronous jobs | `*Batch`, `*Queueable` | `CustomerSyncBatch`, `DocumentArchiveQueueable` |
| Exposed services | `*Resource` | `LeadApiResource` |

### 7.1.3 Classes by functional domain

| Domain | Main classes | Role |
|---|---|---|
| Loans | `LoanApplicationTriggerHandler`, `LoanScoringService`, `LoanApplicationSelector`, `DocumentArchiveQueueable` | Scoring, events, archiving |
| Customer 360 | `Customer360Controller`, `CoreBankingClient`, `CustomerSyncBatch` | Live positions, nightly synchronisation |
| Leads | `LeadApiResource`, `LeadTriggerHandler` | Inbound API, initialisation |
| Notifications | `SmsNotificationService` | SMS reminders |
| Access | `SsoRegistrationHandler` | User creation at single sign-on |

## 7.2 Triggers

| Object | Trigger | Handler | Events |
|---|---|---|---|
| `LoanApplication__c` | `LoanApplicationTrigger` | `LoanApplicationTriggerHandler` | before insert, before update, after update |
| `Lead` | `LeadTrigger` | `LeadTriggerHandler` | before insert |

### 7.2.1 Order of execution

On `Lead`, the trigger runs before save (score initialisation); the flow `Lead_Assign_Branch` runs after save (branch assignment). On `LoanApplication__c`, scoring happens before save, then the event is published and the archiving job is enqueued after update.

## 7.3 Cross-cutting mechanisms

**Loan status propagation.** A single place detects status changes and publishes them; every consequence is a subscriber.

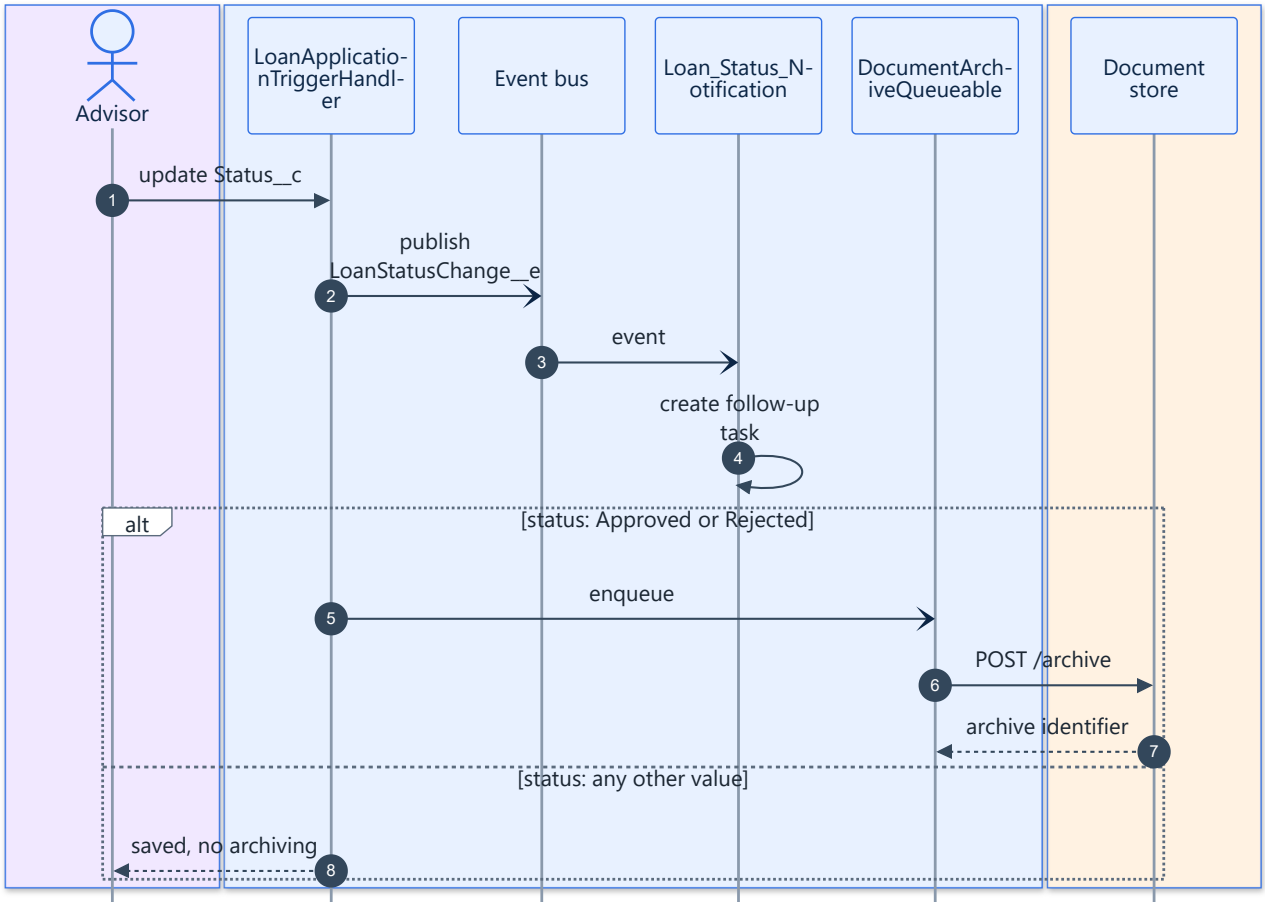



*Figure 11 — Sequence — Propagation of a loan status change*

## 7.4 Asynchronous and scheduled processing

| Job | Type | Trigger or schedule | Purpose |
|---|---|---|---|
| `CustomerSyncBatch` | Batch and Schedulable (batches of 50) | Scheduled (schedule to be confirmed) | Refresh the customer segment from the core banking system |
| `DocumentArchiveQueueable` | Queueable with callouts | Status « Approved » or « Rejected » | Archive the documents of the application |
| `SmsNotificationService.send` | `@future` with callout | Invocable action of `Loan_Document_Reminder` | Send the SMS reminders |

## 7.5 Flows

Flows handle simple record automation and scheduling; Apex handles scoring, integrations and volume.

| Flow | Type | Object | Purpose |
|---|---|---|---|
| `Lead_Assign_Branch` | Record-triggered, after save, on create | `Lead` | Assign the branch of the lead |
| `Loan_Document_Reminder` | Scheduled, daily | `LoanApplication__c` | Remind customers of missing documents by SMS |
| `Loan_Status_Notification` | Platform event-triggered | `LoanStatusChange__e` | Create a follow-up task |

The full list is in Appendix B — Triggers and flows.

## 7.6 OmniStudio

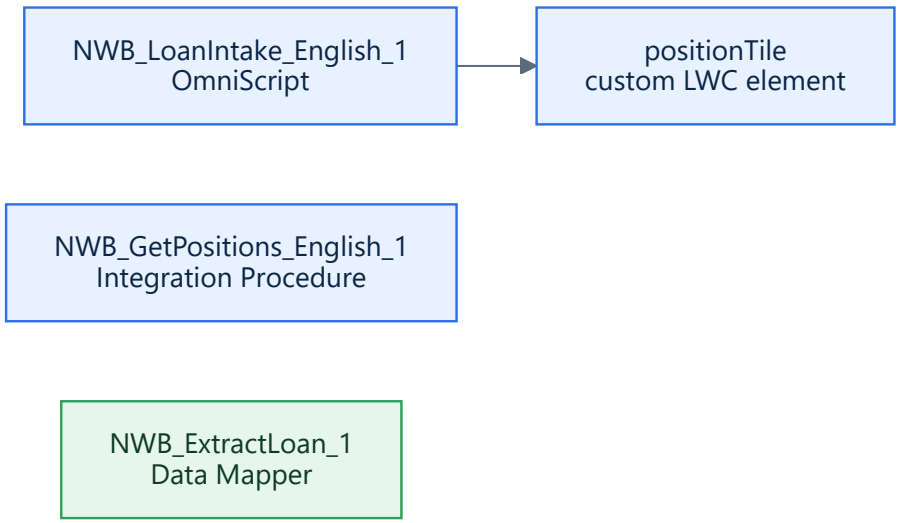



*Figure 12 — OmniStudio components of the loan intake*

| Component | Type | Role |
|---|---|---|
| `NWB_LoanIntake_English_1` | OmniScript (`NWB/LoanIntake`) | Guided loan intake; reuses the `positionTile` component to show the customer's positions |

| Component | Type | Role |
|---|---|---|
| `NWB_GetPositions_English_1` | Integration Procedure | Server-side retrieval of positions (steps to be confirmed) |
| `NWB_ExtractLoan_1` | Data Mapper (extract) | Extraction of loan data (fields to be confirmed) |

## 7.7 Key sequences

**Nightly customer synchronisation.**

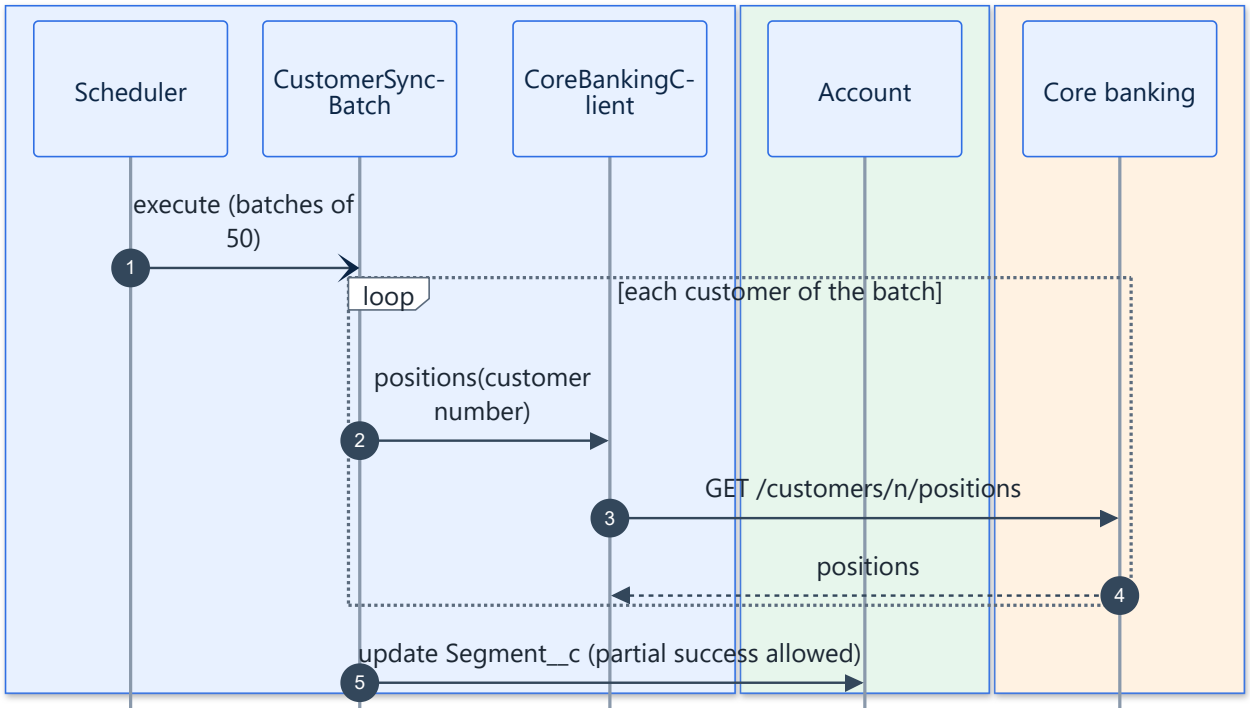



*Figure 13 — Sequence — Nightly synchronisation of the customer segment*

# 8 User interface

## 8.1 Lightning apps and navigation

Both applications are Lightning consoles for desktop:

- « Branch Console » ( `NWB_Branch_Console` ): customers, leads, loan applications, branches and the « New loan » tab; its home page is `Branch_Home` ;
- « Contact Center » ( `NWB_Contact_Center` ): leads, customers and contact attempts.

## 8.2 Lightning pages

| Page | Object | Custom components |
|---|---|---|
| `Account_Record_Page` | `Account` | `customer360` |
| `Loan_Application_Record_Page` | `LoanApplication__c` | `loanTimeline` |
| `Branch_Home` | Home page | Standard components only |

## 8.3 Lightning Web Components

### 8.3.1 By domain

| Domain | Components | Role |
|---|---|---|
| Customer 360 | `customer360` , `positionTile` | Live positions of the customer |
| Loans | `loanTimeline` , `loanWizard` | Timeline of an application; creation wizard in a tab |

## 8.4 Communication between components

- `customer360` reads its data with a cached wire adapter on `Customer360Controller` and passes each position to `positionTile` as a public property.
- `loanTimeline` uses Lightning Message Service; the channel `LoanSelected` is defined for selection events between components.
- `loanWizard` uses the navigation service to open the created application.

# 9 Integration architecture

## 9.1 Integration principles

- The core banking system remains the system of record: Salesforce reads positions live and stores only the keys and the segment.
- Every outbound call uses a named credential; authentication is held by an external credential, never by code.
- Calls from triggers and user transactions are asynchronous (queueable, `@future`); interactive reads are synchronous with an explicit timeout.
- The inbound API creates records in user mode, with the permissions of the integration user.

## 9.2 Integration patterns

| Pattern | Flows |
|---|---|
| Remote Process Invocation — Request and Reply | FLX-01 Customer positions |
| Remote Process Invocation — Fire and Forget | FLX-03 Document archiving, FLX-04 SMS reminders |
| Batch Data Synchronization | FLX-02 Nightly customer synchronisation |
| Remote Call-In | FLX-05 Lead creation from the mobile app |

## 9.3 Flow map

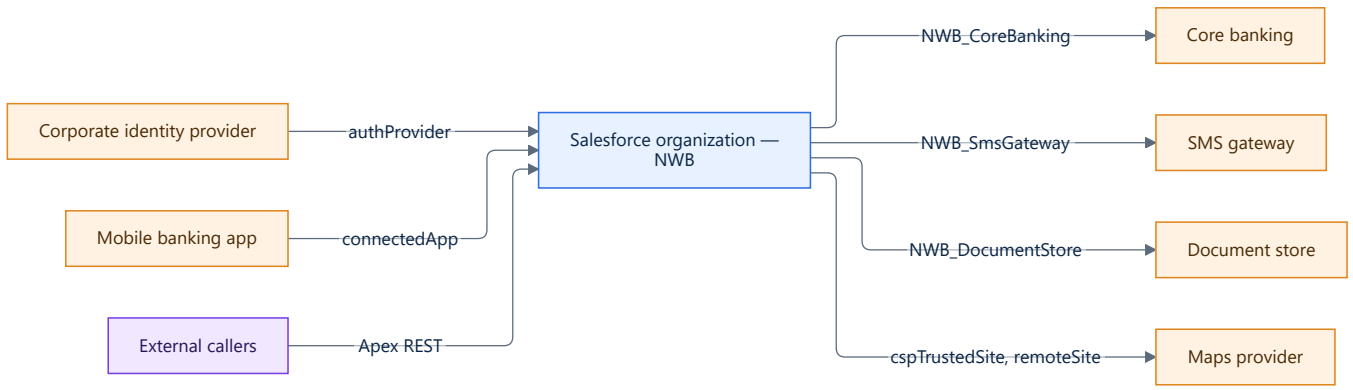



*Figure 14 — Integration map: connected systems and Salesforce endpoints*

| ID | Flow | Direction | System | Salesforce component | Pattern |
|---|---|---|---|---|---|
| FLX-01 | Customer positions | Outbound | Core banking | `CoreBankingClient` | Request and Reply |
| FLX-02 | Customer segment synchronisation | Outbound | Core banking | `CustomerSyncBatch` | Batch Data Synchronization |
| FLX-03 | Document archiving | Outbound | Document store | `DocumentArchiveQueueable` | Fire and Forget |

| ID | Flow | Direction | System | Salesforce component | Pattern |
|---|---|---|---|---|---|
| FLX-04 | SMS reminders | Outbound | SMS gateway | `SmsNotificationService` | Fire and Forget |
| FLX-05 | Lead creation | Inbound | Mobile banking app | `LeadApiResource` | Remote Call-In |
| FLX-06 | Employee login | Inbound | Corporate identity provider | `NWB_Corporate_IdP`, `SsoRegistrationHandler` | Single sign-on |

## 9.4 Connected systems

| System | Description | Salesforce endpoints |
|---|---|---|
| Core banking | System of record for customers, accounts, contracts and positions | `NWB_CoreBanking` |
| SMS gateway | Telecom partner sending SMS notifications | `NWB_SmsGateway` |
| Document store | Long-term archive of the loan documents | `NWB_DocumentStore` |
| Corporate identity provider | Single sign-on for employees (OpenID Connect) | `NWB_Corporate_IdP` |
| Mobile banking app | Customer app that creates leads through the Salesforce API | `NWB_Mobile_Banking` |
| Maps provider | Map tiles and geocoding shown in the branch console | `NWB_Map_Tiles`, `NWB_Maps` |

### 9.4.1 Core banking

Read through `NWB_CoreBanking`, a secured endpoint whose authentication is held by the external credential `NWB_CoreBanking_OAuth` (OAuth, named principal). `CoreBankingClient` calls `GET /customers/{number}/positions` with a 10-second timeout, for the Customer 360 (FLX-01) and the customer synchronisation (FLX-02).

### 9.4.2 Document store

`DocumentArchiveQueueable` posts the identifiers of decided applications to `/archive` through `NWB_DocumentStore`.

### 9.4.3 SMS gateway

`SmsNotificationService` posts messages through `NWB_SmsGateway`, authenticated by the external credential `NWB_SmsGateway_Key`.

### 9.4.4 Maps provider

Map tiles are allowed in the console by the CSP trusted site `NWB_Map_Tiles`; geocoding calls are allowed by the remote site `NWB_Maps`.

## 9.5 Inbound APIs

| URL mapping | Apex class | Nature |
|---|---|---|
| `/leads/v1/*` | `LeadApiResource` | Apex REST |

The mobile banking app authenticates with the connected app `NWB_Mobile_Banking` (scopes `Api` and `RefreshToken`) and calls `POST /services/apexrest/leads/v1/`. The service creates a lead with the name, the channel and, when known, the customer number; the record is inserted with the permissions of the calling user.

## 9.6 Authentication and single sign-on

Employees log in through the authentication provider `NWB_Corporate_IdP` (OpenID Connect). At first login, `SsoRegistrationHandler` creates the user from the identity provider claims and sets its federation identifier.

## 9.7 Allowed domains

| Kind | API name | Host |
|---|---|---|
| Remote Site | `NWB_Maps` | maps.geo-provider.example |
| CSP Trusted Site | `NWB_Map_Tiles` | tiles.geo-provider.example |

Both entries serve the maps provider: tiles in the Lightning pages (CSP) and server-side calls (remote site).

# 10 Security and access

## 10.1 Principles

- Least privilege: a minimal profile, rights added by persona through permission set groups.
- Record access follows the network: private defaults, opened by the role hierarchy and by targeted sharing rules.
- Security is enforced on the server: `with sharing`, user-mode queries, inbound API in user mode.
- No secret in code or configuration files: authentication is held by external credentials.

## 10.2 Authentication and user provisioning

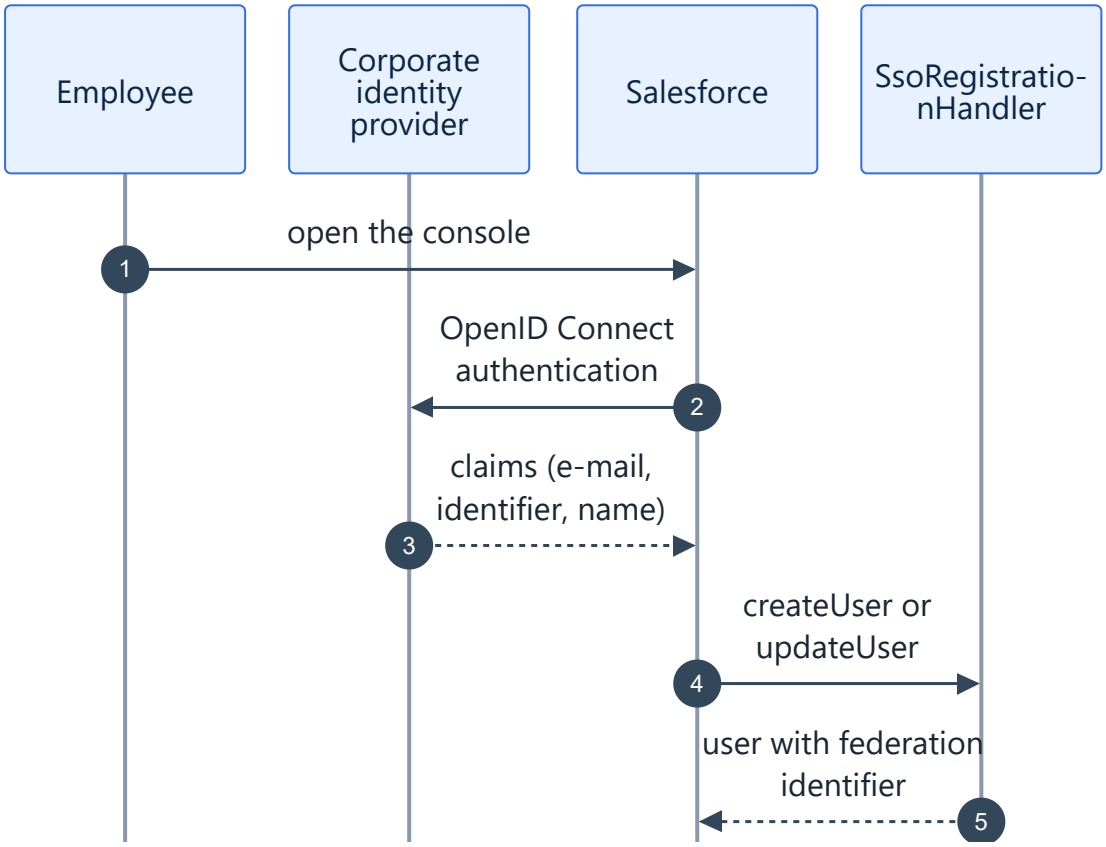



*Figure 15 — Single sign-on and user creation*

The assignment of permission set groups after creation is to be confirmed with the bank (manual or automated).

## 10.3 Personas and permission set groups

| Persona | Permission set group | Main rights |
|---|---|---|
| Branch advisor | `NWB_Persona_Advisor` | `NWB_Advisor`: loan applications and contact attempts (create, edit), branches (read) |

| Persona | Permission set group | Main rights |
|---|---|---|
| Branch manager | `NWB_Persona_Branch_Manager` | `NWB_Advisor` and `NWB_Branch_Manager` : branches (edit) |
| Contact center agent | `NWB_Persona_Agent` | `NWB_Contact_Center_Agent` : contact attempts |

Two technical permission sets complete the model: `NWB_Integration_API` (API access for the integration user) and `NWB_Support_Admin` (support administration).

## 10.4 Role hierarchy

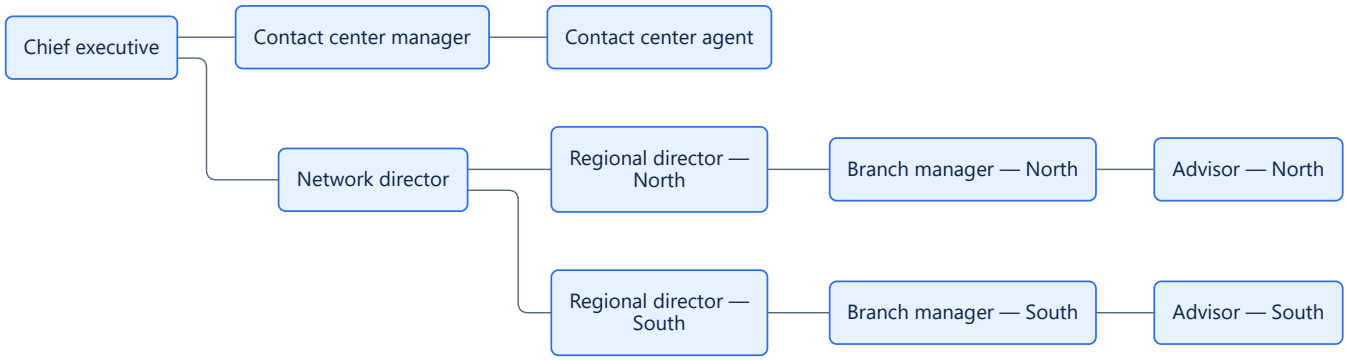



*Figure 16 — Role hierarchy*

The hierarchy mirrors the network: chief executive, network director, regional directors, branch managers and advisors per region, and a separate branch for the contact center.

## 10.5 Sharing model

### 10.5.1 Organization-wide defaults

| Object | Default sharing (internal / external) |
|---|---|
| `Account` | Private / Private |
| `Branch__c` | Read / Private |
| `ContactAttempt__c` | Private / Private |
| `Lead` | Private / Private |
| `LoanApplication__c` | Private / Private |
| `LoanDocument__c` | ControlledByParent / ControlledByParent |
| `Opportunity` | Private / Private |

Customers, leads, opportunities and loan applications are private; branches are readable by all; loan documents follow their application.

### 10.5.2 Sharing rules and queues

| Object | Rule | Kind | Access |
|---|---|---|---|
| `LoanApplication__c` | `Share_Submitted_With_Review_Team` | criteria | Read |

| API name | Label | Object |
|---|---|---|
| `Contact_Center_Leads` | Contact center leads | `Lead` |
| `Loan_Review` | Loan review | `LoanApplication__c` |

Submitted loan applications are opened in read-only mode to the group `All_Branch_Managers`, which includes the managers' superiors. Queues hold the work not yet assigned: loan applications to review and leads for the contact center.

## 10.6 Application security

- 8 classes declare `with sharing` and 1 `inherited sharing`; the nightly synchronisation runs `without sharing` as a system job.
- Queries from the Customer 360 and the selectors run `WITH USER_MODE`; the inbound API inserts as user.
- Validation rules protect data consistency ( `Lead_Locked_When_Converted`, `Amount_Must_Be_Positive` ).

## 10.7 Session settings and protections

| Setting | Value |
|---|---|
| Session timeout | Two hours, with forced logout |
| Session locked to the domain | Yes |
| HttpOnly cookies | Yes |
| Password | 12 characters minimum, upper and lower case, digits and special characters; expires after 90 days; locked after 5 attempts |

## 10.8 Integration users

The mobile banking app calls Salesforce as an integration user with the profile « NWB Integration » (Salesforce Integration licence) and the permission set `NWB_Integration_API`.

# 11 Reporting and analytics

## 11.1 Reports and dashboards

| Folder | Count |
|---|---|
| `Loan_Reports` | 2 |

The folder `Loan_Reports` holds the loan reports by status and by branch; the dashboard `Loan_Pipeline` presents the pipeline to branch managers. The branch home page embeds the dashboards.

# 12 Environments, delivery and operations

## 12.1 Environments

The environments (sandboxes and their purpose) are to be confirmed with the bank. This document describes the production organization.

## 12.2 Release management

The project is a Salesforce DX project (source format, API version 62.0). Metadata is deployed first, OmniStudio components second, then the active versions are checked. The pipeline and its approval gates are to be confirmed.

## 12.3 Scheduled processing

| Processing | Frequency | What to monitor |
|---|---|---|
| `CustomerSyncBatch` | Nightly (to be confirmed) | Failed batches and records not updated (partial success) |
| `Loan_Document_Reminder` | Daily | Flow errors, SMS gateway errors |

## 12.4 Logging and monitoring

Errors of asynchronous jobs are visible in the Apex jobs monitor and flow errors in the flow error e-mails. No specific logging framework is present in the metadata.

## 12.5 Test strategy

5 test classes cover scoring, the trigger handler, the nightly batch and the inbound API. Callouts are isolated behind `CoreBankingClient`, which makes them mockable. Code coverage is to be read in the organization.

# 13 Non-functional requirements

## 13.1 Measures per requirement

| Requirement | Measures in place |
|---|---|
| Security | Single sign-on, persona permission set groups, private defaults, user-mode queries, external credentials |
| Performance | Positions read on demand with a cached wire adapter; callouts with explicit timeouts |
| Volume | Nightly synchronisation in batches of 50 records; asynchronous archiving and SMS |
| Availability | Partial success on synchronisation; Customer 360 independent from the nightly job |
| Maintainability | Layered Apex, naming conventions, configuration in Custom Metadata |
| Traceability | Field history on specific objects, status change events |

## 13.2 Salesforce Well-Architected reading

| Pillar | Strengths observed |
|---|---|
| Trusted | Credentials outside the code, server-side security, least-privilege personas |
| Easy | Two task-oriented consoles, guided intake, live Customer 360 |
| Adaptable | Event-driven follow-up, scoring by configuration, clear separation of layers |

# 14 Architecture decisions

| ID | Decision | Rationale | Consequence |
|---|---|---|---|
| ADR-01 | Read customer positions live from the core banking system | Single source of truth, no duplication of financial data | Customer 360 depends on the availability of the API |
| ADR-02 | Store the scoring rules in Custom Metadata | Business changes without releases | Rules are deployable and auditable like metadata |
| ADR-03 | Publish an event on each loan status change | Add reactions without changing the trigger | Subscribers run asynchronously |
| ADR-04 | Expose an Apex REST service for the mobile app | Control of validation and defaults on the Salesforce side | The service is versioned in its URL ( `/leads/v1/*` ) |

# 15 Outlook

- Extend the event-driven model to the contact center (notifications on lead assignment).
- Enrich the loan intake journey with document upload.
- Open a customer portal for loan follow-up.

The detailed improvement register is kept in the companion Improvement Axes document.

# Appendices

## Appendix A — Apex: asynchronous processing and exposed services

| Apex class | Nature |
|---|---|
| `CustomerSyncBatch` | Batch, Schedulable |
| `DocumentArchiveQueueable` | Callout, Queueable |
| `SmsNotificationService` | Callout, @future |

| URL mapping | Apex class | Nature |
|---|---|---|
| `/leads/v1/*` | `LeadApiResource` | Apex REST |

## Appendix B — Triggers and flows

### B.1 Triggers

| Trigger | Object | Events |
|---|---|---|
| `LeadTrigger` | `Lead` | before insert |
| `LoanApplicationTrigger` | `LoanApplication__c` | before insert, before update, after update |

### B.2 Flows

| API name | Label | Type | Start | Object | Status |
|---|---|---|---|---|---|
| `Lead_Assign_Branch` | Lead — assign branch | AutoLaunchedFlow | RecordAfterSave · Create | `Lead` | Active |
| `Loan_Document_Reminder` | Loan — document reminder | AutoLaunchedFlow | Scheduled · Daily | `LoanApplication__c` | Active |
| `Loan_Status_Notification` | Loan — status notification | AutoLaunchedFlow | PlatformEvent | `LoanStatusChange__e` | Active |

## Appendix C — Custom objects

| API name | Label | Default sharing (internal / external) | Custom fields | Record types | Relationships | Description |
|---|---|---|---|---|---|---|
| `Branch__c` | Branch | Read / Private | 3 | 0 | 1 | Branch of the bank network: code, region and manager. Owner of customers and leads by default. |
| `ContactAttempt__c` | Contact attempt | Private / Private | 3 | 0 | 2 | Outbound contact attempt by an advisor or an agent, with its outcome. |

| API name | Label | Default sharing (internal / external) | Custom fields | Record types | Relationships | Description |
|---|---|---|---|---|---|---|
| `LoanApplication__c` | Loan application | Private / Private | 7 | 2 | 3 | Loan application from intake to decision: customer, amount, score and status. |
| `LoanDocument__c` | Loan document | ControlledByParent / ControlledByParent | 3 | 0 | 1 | — |

### C.1 Extended standard objects

| API name | Custom fields | Record types | Default sharing (internal / external) |
|---|---|---|---|
| `Account` | 4 | 2 | Private / Private |
| `Lead` | 4 | 0 | Private / Private |
| `Opportunity` | 1 | 0 | Private / Private |

### C.2 Validation rules

| Object | Rule | Status |
|---|---|---|
| `Lead` | `Lead_Locked_When_Converted` | Active |
| `LoanApplication__c` | `Amount_Must_Be_Positive` | Active |

## Appendix D — Custom Metadata types

| Type | Records |
|---|---|
| `IntegrationSetting__mdt` | 2 |
| `ScoringRule__mdt` | 3 |

## Appendix E — Active OmniStudio components

### E.1 OmniScripts

| API name | Label | Type | Language | Status |
|---|---|---|---|---|
| `NWB_LoanIntake_English_1` | Loan intake | NWB/LoanIntake | English | Active |

### E.2 Integration Procedures

| API name | Label | Type | Status |
|---|---|---|---|
| `NWB_GetPositions_English_1` | Get positions | NWB/GetPositions | Active |

### E.3 Data Mappers

| API name | Type |
|---|---|
| `NWB_ExtractLoan_1` | Extract |

## Appendix F — User interface

### F.1 Lightning apps

| API name | Label | Navigation | Form factors |
|---|---|---|---|
| `NWB_Branch_Console` | Branch Console | Console | Large |
| `NWB_Contact_Center` | Contact Center | Console | Large |

### F.2 Lightning pages

| API name | Type | Object | Custom components |
|---|---|---|---|
| `Account_Record_Page` | RecordPage | `Account` | `customer360` |
| `Branch_Home` | HomePage | — | — |
| `Loan_Application_Record_Page` | RecordPage | `LoanApplication__c` | `loanTimeline` |

### F.3 Lightning Web Components

| API name | Placed in | Apex controllers |
|---|---|---|
| `customer360` | `Account_Record_Page` | `Customer360Controller` |
| `loanTimeline` | `Loan_Application_Record_Page` | — |
| `loanWizard` | `Loan_Wizard` | — |
| `positionTile` | `customer360`, `NWB_LoanIntake_English_1` | — |

## Appendix G — Access

### G.1 Permission set groups

| API name | Label | Permission sets |
|---|---|---|
| `NWB_Persona_Advisor` | Persona — advisor | `NWB_Advisor` |
| `NWB_Persona_Agent` | Persona — contact center agent | `NWB_Contact_Center_Agent` |
| `NWB_Persona_Branch_Manager` | Persona — branch manager | `NWB_Advisor`, `NWB_Branch_Manager` |

### G.2 Permission sets

| API name | Label | License |
|---|---|---|
| `NWB_Advisor` | Advisor | Salesforce |
| `NWB_Branch_Manager` | Branch manager | Salesforce |
| `NWB_Contact_Center_Agent` | Contact center agent | Salesforce |
| `NWB_Integration_API` | Integration API | Salesforce |
| `NWB_Support_Admin` | Support administration | Salesforce |

### G.3 Profiles

| Name | License |
|---|---|
| `NWB Integration` | Salesforce Integration |
| `NWB Minimum Access` | Salesforce |

### G.4 Roles

| API name | Label | Parent role |
|---|---|---|
| `NWB_Advisor_North` | Advisor — North | `NWB_Branch_Manager_North` |
| `NWB_Advisor_South` | Advisor — South | `NWB_Branch_Manager_South` |
| `NWB_Branch_Manager_North` | Branch manager — North | `NWB_Regional_Director_North` |
| `NWB_Branch_Manager_South` | Branch manager — South | `NWB_Regional_Director_South` |
| `NWB_CEO` | Chief executive | — |
| `NWB_Contact_Center_Agent` | Contact center agent | `NWB_Contact_Center_Manager` |
| `NWB_Contact_Center_Manager` | Contact center manager | `NWB_CEO` |
| `NWB_Network_Director` | Network director | `NWB_CEO` |
| `NWB_Regional_Director_North` | Regional director — North | `NWB_Network_Director` |
| `NWB_Regional_Director_South` | Regional director — South | `NWB_Network_Director` |

## Appendix H — Integration endpoints

### H.1 Named credentials

| API name | Label | Type | Host | External credential |
|---|---|---|---|---|
| `NWB_CoreBanking` | Core banking API | SecuredEndpoint | api.corebanking.northwind-bank.example | `NWB_CoreBanking_OAuth` |
| `NWB_DocumentStore` | Document store | Legacy | docs.northwind-bank.example | — |

| API name | Label | Type | Host | External credential |
|---|---|---|---|---|
| `NWB_SmsGateway` | SMS gateway | SecuredEndpoint | sms.telco-partner.example | `NWB_SmsGateway_Key` |

### H.2 External credentials

| API name | Label | Protocol |
|---|---|---|
| `NWB_CoreBanking_OAuth` | Core banking OAuth (client credentials) | Oauth |
| `NWB_SmsGateway_Key` | SMS gateway API key | Custom |

### H.3 Connected apps and authentication providers

| API name | Label | OAuth scopes |
|---|---|---|
| `NWB_Mobile_Banking` | Mobile banking app | Api, RefreshToken |

| API name | Label | Type | Registration handler |
|---|---|---|---|
| `NWB_Corporate_IdP` | Corporate identity provider | OpenIdConnect | `SsoRegistrationHandler` |

### H.4 Allowed domains

| Kind | API name | Host |
|---|---|---|
| Remote Site | `NWB_Maps` | maps.geo-provider.example |
| CSP Trusted Site | `NWB_Map_Tiles` | tiles.geo-provider.example |

## Appendix I — Packages and namespaces

| Namespace | Product |
|---|---|
| `FinServ` | Financial Services Cloud |

## Appendix J — Traceability to the metadata

| Chapter | Metadata analysed |
|---|---|
| 4 General architecture | Applications, Lightning pages, installed packages and namespaces |
| 6 Data architecture | Objects, fields, relationships, record types, validation rules, Custom Metadata |
| 7 Application architecture | Apex classes and triggers, flows, OmniStudio components |
| 8 User interface | Applications, Lightning pages, LWC, Aura, Visualforce, tabs, quick actions |
| 9 Integration architecture | Named and External Credentials, remote sites, connected apps, auth providers, CSP and CORS, Apex REST |
| 10 Security and access | Profiles, permission sets and groups, roles, queues, sharing rules, security settings |

| Chapter | Metadata analysed |
|---|---|
| 12 Environments, delivery and operations | Scheduled jobs, test classes, code coverage (when connected to the org) |

| Type | Count |
|---|---|
| `applications` | 2 |
| `authproviders` | 1 |
| `classes` | 18 |
| `connectedApps` | 1 |
| `cspTrustedSites` | 1 |
| `customMetadata` | 5 |
| `dashboards` | 1 |
| `email` | 1 |
| `externalCredentials` | 2 |
| `flexipages` | 3 |
| `flows` | 4 |
| `groups` | 1 |
| `labels` | 1 |
| `lwc` | 5 |
| `messageChannels` | 1 |
| `namedCredentials` | 3 |
| `objects` | 46 |
| `omniDataTransforms` | 1 |
| `omniIntegrationProcedures` | 1 |
| `omniScripts` | 1 |
| `permissionsetgroups` | 3 |
| `permissionsets` | 5 |
| `profiles` | 2 |
| `queues` | 2 |
| `quickActions` | 1 |
| `remoteSiteSettings` | 1 |
| `reports` | 3 |
| `roles` | 10 |
| `settings` | 1 |
| `sharingRules` | 1 |

| Type | Count |
|---|---|
| `staticresources` | 1 |
| `tabs` | 4 |
| `triggers` | 2 |

## Appendix K — Glossary

| Term | Definition |
|---|---|
| Branch | Unit of the bank network that owns customers and leads by default (Branch__c). |
| Custom Metadata | Deployable configuration records, read by code and flows without consuming queries. |
| External Credential | Authentication definition (protocol, principals) used by named credentials. |
| Flow | Declarative automation of the Salesforce platform (record-triggered, scheduled, screen, autolaunched). |
| loan application | Request for a consumer loan or a mortgage, from intake to decision (LoanApplication__c). |
| LWC | Lightning Web Component: standards-based UI component of the Lightning platform. |
| Named Credential | Secure definition of a callout endpoint and its authentication. |
| OWD | Organization-Wide Defaults: baseline record access for each object. |
| PSG | Permission Set Group: bundle of permission sets assigned as one persona. |
| Queue | Shared owner for records awaiting assignment. |
| Record Type | Variant of an object with its own business process, picklist values and page layout. |